VERIFICATION CORE REPORT

Topic: **Java 21 Virtual Threads & Structured Concurrency** | College: **DPU**
Tech: **Java 21** | Professor: **Dr. S. Walanju** | Deck ID: **66df3f28** | Date: 2026-10-17 13:16:36
Docker Image Digest: **eclipse-temurin:21-jdk-alpine@sha256:7b5d1e4392a874f6e21b03e2c34a6e8b7c439f018e652a129efb923415cf98a1**

Metric	Value	Metric	Value
Found vs Executed	16 found / 16 run	Unverified (Not Run)	0
Snippets Verified	0/16	Auto-Healed	0
Needs Review / Fail	15	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	9333%
Template QA Passed	10/10	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	java 21 (isolated_local_subprocess)	NEEDS REVIEW	320.96910476684576ms	(none)
Slide 4	java 21 (isolated_local_subprocess)	NEEDS REVIEW	321.75683975219727ms	(none)
Slide 5	java 21 (isolated_local_subprocess)	NEEDS REVIEW	323.74596595764166ms	(none)
Slide 6	java 21 (isolated_local_subprocess)	NEEDS REVIEW	325.98209381103516ms	(none)
Slide 7	java 21 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	494.05043472290044ms	`Result: Solution Verified`
Slide 8	java 21 (isolated_local_subprocess)	NEEDS REVIEW	326.09176635742188ms	(none)
Slide 3	java 21.0.6 (isolated_local_subprocess)	NEEDS REVIEW	333.25266838073730ms	(none)
Slide 4	java 21.0.6 (isolated_local_subprocess)	NEEDS REVIEW	328.7518024444580ms	(none)
Slide 5	java 21.0.6 (isolated_local_subprocess)	NEEDS REVIEW	371.94657325744630ms	(none)

Slide 6	java 21.0.6 (isolated_local_subprocess)	NEEDS REVIEW	330.5649757385254ms	(none)
Slide 8	java 21.0.6 (isolated_local_subprocess)	NEEDS REVIEW	338.4933471679687ms	(none)
Slide 3	java 21 (isolated_local_subprocess)	NEEDS REVIEW	330.6150436401367ms	(none)
Slide 4	java 21 (isolated_local_subprocess)	NEEDS REVIEW	333.9605331420898ms	(none)
Slide 5	java 21 (isolated_local_subprocess)	NEEDS REVIEW	397.8497982025146ms	(none)
Slide 6	java 21 (isolated_local_subprocess)	NEEDS REVIEW	338.6824131011963ms	(none)
Slide 8	java 21 (isolated_local_subprocess)	NEEDS REVIEW	335.2611064910888ms	(none)

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-7-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	4/5	80.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-7-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.